

# Exponentials & Logarithms

## ANSWER KEY



### Evaluating exponential expressions

- 1 Evaluate  $2^5$ .

$$2^5 = 32.$$

- 2 Evaluate  $3^{-2}$ .

$$3^{-2} = \frac{1}{9}.$$

- 3 Evaluate  $16^{1/2}$ .

$$16^{1/2} = 4.$$

- 4 Evaluate  $27^{2/3}$ .

$$27^{2/3} = (27^{1/3})^2 = 3^2 = 9.$$

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### Evaluating logarithms

- 5 Evaluate  $\log_2 32$ .

$$\text{Since } 2^5 = 32, \log_2 32 = 5.$$

- 6 Evaluate  $\log_3 81$ .

$$\text{Since } 3^4 = 81, \log_3 81 = 4.$$

- 7 Evaluate  $\log_5 1$ .

$$\log_5 1 = 0.$$

- 8 Evaluate  $\log_{10} 0.01$ .

$$\text{Since } 10^{-2} = 0.01, \log_{10} 0.01 = -2.$$

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### Log laws: product and quotient

- 9 Write  $\log_2 (8x)$  as a sum of logarithms.

$$\log_2 (8x) = \log_2 8 + \log_2 x = 3 + \log_2 x.$$

- 10 Write  $\log_3 \left( \frac{x}{9} \right)$  as a difference of logarithms.

$$\log_3 \left( \frac{x}{9} \right) = \log_3 x - \log_3 9 = \log_3 x - 2.$$

- 11 Combine  $\log_5 5 + \log_5 25$  into a single logarithm and evaluate.

$$\log_5 (5 \times 25) = \log_5 125 = 3 \text{ (since } 5^3 = 125 \text{)}.$$

- 12 Combine  $\log_2 24 - \log_2 3$  into a single logarithm and evaluate.

$$\log_2 \left( \frac{24}{3} \right) = \log_2 8 = 3.$$

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### Log laws: the power rule

- 13 Write  $\log(x^5)$  using the power rule.

$$\log(x^5) = 5\log x.$$

- 14 Evaluate  $\log_2 (4^3)$  using the power rule.

$$\log_2 (4^3) = 3\log_2 4 = 3(2) = 6.$$

- 15 Write  $2\log_3 x + \log_3 y$  as a single logarithm.

$$2\log_3 x + \log_3 y = \log_3 (x^2 y).$$

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### Expanding and condensing logarithmic expressions

- 16 Expand  $\log \left( \frac{x^3 y}{z^2} \right)$  fully.

$$3\log x + \log y - 2\log z.$$

- 17 Condense  $3\ln x - 2\ln y$  into a single logarithm.

$$\ln \left( \frac{x^3}{y^2} \right).$$

- 18 Expand  $\log_2 (\sqrt{x} \cdot y^3)$  fully.

$$\sqrt{x} = x^{1/2}, \text{ so the expression equals } \frac{1}{2}\log_2 x + 3\log_2 y.$$

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### Solving exponential equations (matching bases)

- 19 Solve  $2^x = 32$ .

$$32 = 2^5, \text{ so } x = 5.$$

20 Solve  $3^{x+1} = 81$ .

$$81 = 3^4, \text{ so } x + 1 = 4 \Rightarrow x = 3.$$

21 Solve  $5^{2x} = 125$ .

$$125 = 5^3, \text{ so } 2x = 3 \Rightarrow x = \frac{3}{2}.$$

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### Solving exponential equations using logarithms

22 Solve  $2^x = 10$  for  $x$ , rounded to three decimal places.

$$x = \frac{\log 10}{\log 2} \approx \frac{1}{0.301} \approx 3.322.$$

23 Solve  $5^x = 40$  for  $x$ , rounded to three decimal places.

$$x = \frac{\log 40}{\log 5} \approx \frac{1.602}{0.699} \approx 2.292.$$

24 Solve  $3^{2x} = 50$  for  $x$ , rounded to three decimal places.

$$2x = \frac{\log 50}{\log 3} \approx \frac{1.699}{0.477} \approx 3.561, \text{ so } x \approx 1.781.$$

25 Solve  $e^x = 15$  for  $x$ , rounded to three decimal places.

$$x = \ln 15 \approx 2.708.$$

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### Solving logarithmic equations

26 Solve  $\log_2(x) = 5$ .

$$x = 2^5 = 32.$$

27 Solve  $\log_3(x - 1) = 2$ .

$$x - 1 = 3^2 = 9 \Rightarrow x = 10.$$

28 Solve  $\log(x) + \log(x - 3) = 1$ .

$$\log[x(x - 3)] = 1 \Rightarrow x(x - 3) = 10 \Rightarrow x^2 - 3x - 10 = 0 \Rightarrow (x - 5)(x + 2) = 0. \text{ Reject } x = -2 \text{ (logarithm undefined); } x = 5.$$

29 Solve  $\log_2(x + 3) + \log_2(x - 1) = 5$ .

$$\log_2[(x + 3)(x - 1)] = 5 \Rightarrow (x + 3)(x - 1) = 32 \Rightarrow x^2 + 2x - 35 = 0 \Rightarrow (x + 7)(x - 5) = 0. \text{ Reject } x = -7 \text{ (makes } x - 1 < 0); x = 5.$$

## Natural logarithms and e

- 30 Evaluate  $\ln(e^3)$ .

$$\ln(e^3) = 3.$$

- 31 Evaluate  $\ln(1)$ .

$$\ln(1) = 0.$$

- 32 Solve  $\ln(x) = 2$  for  $x$ , in terms of  $e$ .

$$x = e^2.$$

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## Exponential growth and decay models

- 33 The population of a town is modeled by  $P(t) = 5000e^{0.03t}$ , where  $t$  is in years. Find  $P(10)$ , rounded to the nearest whole number.

$$P(10) = 5000e^{0.3} = 5000(1.34986) \approx 6749.$$

- 34 A radioactive substance decays according to  $A(t) = 200e^{-0.05t}$  (grams). Find  $A(20)$ , rounded to one decimal place.

$$A(20) = 200e^{-1} = 200(0.36788) \approx 73.6 \text{ grams.}$$

- 35 Determine whether  $N(t) = 50e^{0.02t}$  models growth or decay, and justify your answer.

Growth — the exponent's coefficient 0.02 is positive, so  $N$  increases as  $t$  increases.

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## Word problems: exponential models in context

- 36 This question has two parts. An investment of 2000 dollars grows according to  $A(t) = 2000e^{0.04t}$ , where  $t$  is in years.

**a** Find the value of the investment after 5 years, rounded to the nearest dollar.

$$A(5) = 2000e^{0.2} = 2000(1.2214) \approx 2443 \text{ dollars.}$$

**b** Find how many years it takes for the investment to double, rounded to one decimal place (using  $\ln 2 \approx 0.693$ ).  $2 = e^{0.04t} \Rightarrow 0.04t = \ln 2 \approx 0.693 \Rightarrow t \approx 17.3$  years.

- 37** This question has two parts. A cup of coffee cools according to  $T(t) = 20 + 60e^{-0.1t}$  ( $^{\circ}\text{C}$ ), where  $t$  is minutes after brewing and  $20^{\circ}\text{C}$  is room temperature.
- a** Find the temperature after 15 minutes, rounded to one decimal place (using  $e^{-1.5} \approx 0.2231$ ).  
 $T(15) = 20 + 60e^{-1.5} = 20 + 60(0.2231) \approx 33.4^{\circ}\text{C}$ .
- b** Find how many minutes it takes for the temperature to drop to  $35^{\circ}\text{C}$ , rounded to one decimal place (using  $\ln 4 \approx 1.386$ ).  $35 = 20 + 60e^{-0.1t} \Rightarrow e^{-0.1t} = 0.25 \Rightarrow -0.1t = -\ln 4 \approx -1.386 \Rightarrow t \approx 13.9$  minutes.
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### MCQ — Evaluating exponential expressions

- 38** Evaluate  $3^4$ .

- a.** 81 ✓
- b.** 243
- c.** 12
- d.** 64

$$3^4 = 81.$$

- 39** Evaluate  $4^{-2}$ .

- a.**  $-\frac{1}{16}$
- b.**  $\frac{1}{16}$  ✓
- c.**  $\frac{1}{8}$
- d.** 16

$$4^{-2} = \frac{1}{4^2} = \frac{1}{16}.$$

40 Evaluate  $25^{1/2}$ .

a. 5 ✓

b. 50

c. 12.5

d. 625

$$25^{1/2} = \sqrt{25} = 5.$$

41 Evaluate  $8^{2/3}$ .

a. 5.33

b. 4 ✓

c. 2

d. 16

$$8^{2/3} = (8^{1/3})^2 = 2^2 = 4.$$

42 Evaluate  $2^{-3}$ .

a. -8

b.  $\frac{1}{6}$

c. 8

d.  $\frac{1}{8}$  ✓

$$2^{-3} = \frac{1}{2^3} = \frac{1}{8}.$$

43 Evaluate  $32^{3/5}$ .

a. 2

b. 16

c. 8 ✓

d. 32

$$32^{3/5} = (32^{1/5})^3 = 2^3 = 8.$$

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## MCQ — Evaluating logarithms

44 Evaluate  $\log_3 27$ .

a. 3 ✓

b. 1

c. 27

d. 9

$$3^3 = 27.$$

45 Evaluate  $\log_2 64$ .

a. 32

b. 8

c. 5

d. 6 ✓

$$2^6 = 64.$$

46 Evaluate  $\log_7 1$ .

a. 1

b. Undefined

c. 7

d. 0 ✓

$$7^0 = 1.$$

47 Evaluate  $\log_{10} 0.001$ .

a. 0.001

b.  $-3$  ✓

c.  $-2$

d. 3

$$10^{-3} = 0.001.$$

48 Evaluate  $\log_4 16$ .

a. 4

b. 8

c. 2 ✓

d. 16

$$4^2 = 16.$$

49 Evaluate  $\log_5 125$ .

a. 5

b. 3 ✓

c. 1

d. 25

$$5^3 = 125.$$

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### MCQ — Log laws: product and quotient



50 Write  $\log_3 (9x)$  as a sum of logarithms.

a.  $2 + \log_3 x$  ✓

b.  $2\log_3 x$

c.  $9 + \log_3 x$

d.  $\log_3 9 + \log_3 x$  only

$$\log_3 (9x) = \log_3 9 + \log_3 x = 2 + \log_3 x.$$

51 Write  $\log_5 \left( \frac{x}{25} \right)$  as a difference of logarithms.

a.  $\log_5 x - 25$

b.  $2 - \log_5 x$

c.  $\log_5 x - 2$  ✓

d.  $\frac{\log_5 x}{2}$

$$\log_5 \left( \frac{x}{25} \right) = \log_5 x - \log_5 25 = \log_5 x - 2.$$

52 Combine  $\log_2 4 + \log_2 8$  into a single logarithm and evaluate.

a. 32

b. 5 ✓

c. 12

d. 2

$$\log_2 (4 \times 8) = \log_2 32 = 5.$$

53 Combine  $\log_3 54 - \log_3 2$  into a single logarithm and evaluate.

a. 27

b. 3 ✓

c. 26

d. 52

$$\log_3 (54 \div 2) = \log_3 27 = 3.$$

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## MCQ — Log laws: the power rule

54 Write  $\log(x^7)$  using the power rule.

- a.  $7\log x$  ✓
- b.  $x^7\log x$
- c.  $\log x^7$  unchanged
- d.  $\log(7x)$

Power rule:  $\log(x^n) = n\log x$ .

55 Evaluate  $\log_3(9^2)$  using the power rule.

- a. 18
- b. 81
- c. 2
- d. 4 ✓

$$2\log_3 9 = 2(2) = 4.$$

56 Write  $3\log_2 x + \log_2 y$  as a single logarithm.

- a.  $3\log_2(xy)$
- b.  $\log_2(x^3 + y)$
- c.  $\log_2(x^3y)$  ✓
- d.  $\log_2(3xy)$

$3\log_2 x = \log_2(x^3)$ ; combined with  $\log_2 y$ :  $\log_2(x^3y)$ .

57 Write  $4\log x - \log y$  as a single logarithm.

a.  $\log\left(\frac{x^4}{y}\right)$  ✓

b.  $\log(4x - y)$

c.  $4\log\left(\frac{x}{y}\right)$

d.  $\log\left(\frac{4x}{y}\right)$

$4\log x = \log(x^4)$ ; combined:  $\log\left(\frac{x^4}{y}\right)$ .

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### MCQ — Expanding and condensing logarithmic expressions

58 Expand  $\log\left(\frac{x^2y}{z^3}\right)$  fully.

a.  $2\log x + \log y - 3\log z$  ✓

b.  $2\log x - \log y - 3\log z$

c.  $\log x^2 + \log y - \log z^3$  unchanged

d.  $2\log x + \log y + 3\log z$

Apply the quotient and power rules:  $2\log x + \log y - 3\log z$ .

59 Condense  $2\ln x - 3\ln y$  into a single logarithm.

a.  $\ln\left(\frac{x^2}{y^3}\right)$  ✓

b.  $\ln\left(\frac{x^2}{y}\right) - 3$

c.  $\ln\left(\frac{x^3}{y^2}\right)$

d.  $\ln(x^2 - y^3)$

$2\ln x = \ln(x^2)$ ,  $3\ln y = \ln(y^3)$ :  $\ln\left(\frac{x^2}{y^3}\right)$ .

60 Expand  $\log_2 (\sqrt{x} \cdot y^4)$  fully.

a.  $\frac{1}{2}\log_2 x + 4\log_2 y$  ✓

b.  $\frac{1}{2}\log_2 x + \log_2 y^4$  unchanged

c.  $2\log_2 x + 4\log_2 y$

d.  $\frac{1}{4}\log_2 x + 2\log_2 y$

$\sqrt{x} = x^{1/2}$ :  $\frac{1}{2}\log_2 x + 4\log_2 y$ .

61 Condense  $\frac{1}{2}\log x + 2\log y$  into a single logarithm.

a.  $\log(x \cdot y^2)$

b.  $\log\left(\frac{\sqrt{x}}{y^2}\right)$

c.  $\log(\sqrt{x} \cdot y^2)$  ✓

d.  $\log(2\sqrt{x} \cdot y)$

$\frac{1}{2}\log x = \log \sqrt{x}$ ; combined:  $\log(\sqrt{x} \cdot y^2)$ .

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### MCQ — Solving exponential equations (matching bases)

62 Solve  $2^x = 64$ .

a.  $x = 8$

b.  $x = 6$  ✓

c.  $x = 32$

d.  $x = 5$

$2^6 = 64$ .

63 Solve  $3^{x+2} = 27$ .

a.  $x = 9$

b.  $x = 1$  ✓

c.  $x = 5$

d.  $x = 3$

$$3^{x+2} = 3^3 \Rightarrow x + 2 = 3 \Rightarrow x = 1.$$

64 Solve  $5^{3x} = 125$ .

a.  $x = \frac{1}{3}$

b.  $x = 41.67$

c.  $x = 3$

d.  $x = 1$  ✓

$$5^{3x} = 5^3 \Rightarrow 3x = 3 \Rightarrow x = 1.$$

65 Solve  $4^x = \frac{1}{16}$ .

a.  $x = 4$

b.  $x = -4$

c.  $x = -2$  ✓

d.  $x = 2$

$$4^x = 4^{-2} \Rightarrow x = -2.$$

66 Solve  $2^{2x-1} = 32$ .

a.  $x = 3$  ✓

b.  $x = 2.5$

c.  $x = 6$

d.  $x = 5.5$

$$2^{2x-1} = 2^5 \Rightarrow 2x - 1 = 5 \Rightarrow x = 3.$$

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## MCQ — Solving exponential equations using logarithms

67 Solve  $2^x = 12$  for  $x$ , rounded to three decimal places.

a. 3.585 ✓

b. 3.170

c. 4.000

d. 2.485

$$x = \frac{\ln 12}{\ln 2} \approx 3.585.$$

68 Solve  $7^x = 50$  for  $x$ , rounded to three decimal places.

a. 7.143

b. 2.010 ✓

c. 3.000

d. 1.505

$$x = \frac{\ln 50}{\ln 7} \approx 2.010.$$

69 Solve  $4^{2x} = 90$  for  $x$ , rounded to three decimal places.

a. 6.492

b. 3.246

c. 1.623 ✓

d. 0.811

$$2x = \frac{\ln 90}{\ln 4} \approx 3.246 \Rightarrow x \approx 1.623.$$

70 Solve  $e^x = 25$  for  $x$ , rounded to three decimal places.

- a. 25.000
- b. 2.303
- c. 3.219 ✓
- d. 12.500

$$x = \ln 25 \approx 3.219.$$

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### MCQ — Solving logarithmic equations

71 Solve  $\log_3(x) = 4$ .

- a.  $x = 7$
- b.  $x = 12$
- c.  $x = 64$
- d.  $x = 81$  ✓

$$3^4 = 81.$$

72 Solve  $\log_2(x - 1) = 3$ .

- a.  $x = 6$
- b.  $x = 9$  ✓
- c.  $x = 7$
- d.  $x = 8$

$$x - 1 = 2^3 = 8 \Rightarrow x = 9.$$

73 Solve  $\log(x) + \log(x - 3) = 1$ .

a.  $x = -2$

b.  $x = 5$  ✓

c.  $x = 10$

d.  $x = 13$

$\log(x(x - 3)) = 1 \Rightarrow x^2 - 3x - 10 = 0 \Rightarrow (x - 5)(x + 2) = 0$ ; only  $x = 5$  keeps both logs defined.

74 Solve  $\log_2(x + 2) + \log_2(x - 2) = 5$ .

a.  $x = 6$  ✓

b.  $x = -6$

c.  $x = 34$

d.  $x = 18$

$(x + 2)(x - 2) = 2^5 = 32 \Rightarrow x^2 - 4 = 32 \Rightarrow x^2 = 36$ ; only  $x = 6$  keeps both logs defined.

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### MCQ — Natural logarithms and e

75 Evaluate  $\ln(e^5)$ .

a. 0

b.  $e^5$

c. 1

d. 5 ✓

$\ln(e^x) = x$ .



76 Evaluate  $\ln\left(\frac{1}{e}\right)$ .

a. 1

b.  $e$

c.  $-1$  ✓

d. 0

$$\ln(e^{-1}) = -1.$$

77 Solve  $\ln(x) = 3$  for  $x$ , in terms of  $e$ .

a.  $x = e^{1/3}$

b.  $x = 3$

c.  $x = e^3$  ✓

d.  $x = 3e$

$$\ln(x) = 3 \Rightarrow x = e^3.$$

78 Evaluate  $e^{\ln 7}$ .

a. 7 ✓

b.  $\ln 7$

c. 1

d.  $e^7$

$$e^{\ln x} = x.$$

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## MCQ — Exponential growth and decay models

- 79 The population of a town is modeled by  $P(t) = 8,000e^{0.025t}$ , where  $t$  is in years. Find  $P(15)$ , rounded to the nearest whole number.

- a. 8,300
- b. 11,000
- c. 11,640 ✓
- d. 12,000

$$P(15) = 8,000e^{0.375} \approx 11,640.$$

- 80 A radioactive substance decays according to  $A(t) = 150e^{-0.04t}$  (grams). Find  $A(25)$ , rounded to one decimal place.

- a. 60.7
- b. 50.0
- c. 75.0
- d. 55.2 ✓

$$A(25) = 150e^{-1} \approx 55.2 \text{ g.}$$

- 81 Determine whether  $N(t) = 80e^{-0.015t}$  models growth or decay.

- a. Decay ✓
- b. Neither
- c. Growth
- d. Cannot be determined

The exponent's coefficient is negative: decay.

82 Determine whether  $M(t) = 30e^{0.01t}$  models growth or decay.

- a. Neither
- b. Cannot be determined
- c. Decay
- d. Growth ✓

The exponent's coefficient is positive: growth.

83 The graph shows an exponential function. Based on the graph, does it represent growth or decay?

- a. Neither
- b. Growth ✓
- c. Cannot be determined
- d. Decay

The curve rises as  $x$  increases: growth.

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### MCQ — Word problems: exponential models in context

84 An investment of 3,000 dollars grows according to  $A(t) = 3,000e^{0.05t}$  dollars, where  $t$  is in years. Find the investment's value after 8 years, rounded to the nearest dollar.

- a. 4,200
- b. 4,800
- c. 4,475 ✓
- d. 3,400

$$A(8) = 3,000e^{0.4} \approx 4,475 \text{ dollars.}$$

- 85 A cup of coffee cools according to  $T(t) = 22 + 70e^{-0.08t}$  ( $^{\circ}\text{C}$ ), where  $t$  is in minutes. Find the temperature after 15 minutes, rounded to one decimal place.

- a.  $38.5^{\circ}\text{C}$
- b.  $50.0^{\circ}\text{C}$
- c.  $43.1^{\circ}\text{C}$  ✓
- d.  $22.0^{\circ}\text{C}$

$$T(15) = 22 + 70e^{-1.2} \approx 43.1^{\circ}\text{C}.$$

- 86 A population of bacteria doubles every 3 hours, starting at 500. Using  $P(t) = 500 \cdot 2^{t/3}$ , find the population after 9 hours.

- a. 4,000 ✓
- b. 2,000
- c. 8,000
- d. 1,500

$$P(9) = 500 \times 2^3 = 4,000.$$

- 87 The value of a car depreciates according to  $V(t) = 25,000(0.85)^t$  dollars, where  $t$  is in years. Find the car's value after 5 years, rounded to the nearest dollar.

- a. 11,093 ✓
- b. 12,750
- c. 9,500
- d. 10,625

$$V(5) = 25,000(0.85)^5 \approx 11,093 \text{ dollars}.$$

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**Multiple choice — easy**

88 Evaluate  $5^{-3}$ .

a.  $\frac{1}{125}$  ✓

b.  $-\frac{1}{125}$

c. 125

d.  $\frac{1}{127}$

$$5^{-3} = \frac{1}{125}.$$

89 Evaluate  $4^{\frac{1}{2}}$ .

a. 4

b. 5

c. 2 ✓

d. 6

$$4^{\frac{1}{2}} = 2.$$

90 Evaluate  $5^{-2}$ .

a. 25

b.  $\frac{1}{27}$

c.  $-\frac{1}{25}$

d.  $\frac{1}{25}$  ✓

$$5^{-2} = \frac{1}{25}.$$

91 Evaluate  $5^2$ .

a. 27

b. 10

c. 23

d. 25 ✓

$$5^2 = 25.$$

92 Evaluate  $3^{-2}$ .

a.  $\frac{1}{11}$

b.  $-\frac{1}{9}$

c.  $\frac{1}{9}$  ✓

d. 9

$$3^{-2} = \frac{1}{9}.$$

93 Evaluate  $\log_2 4$ .

a. 3

b. 1

c. 2 ✓

d. 4

$$\text{Since } 2^2 = 4, \log_2 4 = 2.$$

94 Evaluate  $\log_4 1$ .

a. 0 ✓

b. 4

c. -1

d. 1

$$\log_4 1 = 0.$$

95 Evaluate  $\log_{10} 0.01$ .

- a. 2
- b.  $-2$  ✓
- c.  $-3$
- d.  $-1$

Since  $10^{-2} = 0.01$ ,  $\log_{10} 0.01 = -2$ .

96 Evaluate  $\log_{10} 0.1$ .

- a.  $-2$
- b.  $-1$  ✓
- c.  $-3$
- d. 1

Since  $10^{-1} = 0.1$ ,  $\log_{10} 0.1 = -1$ .

97 Evaluate  $\log_5 625$ .

- a. 625
- b. 3
- c. 5
- d. 4 ✓

Since  $5^4 = 625$ ,  $\log_5 625 = 4$ .

98 Solve  $4^x = 256$ .

- a. 4 ✓
- b. 3
- c. 6
- d. 5

$256 = 4^4$ , so  $x = 4$ .

99 Solve  $3^{x+3} = 81$ .

a. 4

b. 1 ✓

c. 2

d. 0

$81 = 3^4$ , so  $x + 3 = 4 \Rightarrow x = 1$ .

100 Solve  $4^{x+3} = 64$ .

a. 1

b. -1

c. 0 ✓

d. 3

$64 = 4^3$ , so  $x + 3 = 3 \Rightarrow x = 0$ .

101 Solve  $5^x = 125$ .

a. 5

b. 3 ✓

c. 2

d. 4

$125 = 5^3$ , so  $x = 3$ .



102 Solve  $4^{x-3} = 16$ .

a. 5 ✓

b. 6

c. 2

d. 4

$16 = 4^2$ , so  $x - 3 = 2 \Rightarrow x = 5$ .

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### Multiple choice — medium

103 Combine  $\log_2 2 + \log_2 2$  into a single logarithm and evaluate.

a. 3

b. 1

c. 4

d. 2 ✓

$\log_2 (2 \times 2) = \log_2 4 = 2$  (since  $2^2 = 4$ ).

104 Combine  $\log_5 125 + \log_5 125$  into a single logarithm and evaluate.

a. 7

b. 5

c. 15625

d. 6 ✓

$\log_5 (125 \times 125) = \log_5 15625 = 6$  (since  $5^6 = 15625$ ).

**105** Combine  $\log_5 125 - \log_5 25$  into a single logarithm and evaluate.

a. 3

b. 5

c. 1 ✓

d. 2

$$\log_5 \left( \frac{125}{25} \right) = \log_5 5 = 1.$$

**106** Combine  $\log_5 5 + \log_5 25$  into a single logarithm and evaluate.

a. 125

b. 2

c. 3 ✓

d. 4

$$\log_5 (5 \times 25) = \log_5 125 = 3 \text{ (since } 5^3 = 125 \text{)}.$$

**107** Combine  $\log_5 125 + \log_5 25$  into a single logarithm and evaluate.

a. 5 ✓

b. 6

c. 3125

d. 4

$$\log_5 (125 \times 25) = \log_5 3125 = 5 \text{ (since } 5^5 = 3125 \text{)}.$$

**108** Evaluate  $\log_2 (8^2)$  using the power rule.

a. 6 ✓

b. 8

c. 5

d. 4

$$\log_2 (8^2) = 2\log_2 8 = 2(3) = 6.$$

**109** Evaluate  $\log_2 (4^4)$  using the power rule.

a. 6

b. 8 ✓

c. 12

d. 4

$$\log_2 (4^4) = 4\log_2 4 = 4(2) = 8.$$

**110** Evaluate  $\log_3 (9^4)$  using the power rule.

a. 4

b. 8 ✓

c. 12

d. 6

$$\log_3 (9^4) = 4\log_3 9 = 4(2) = 8.$$

**111** Evaluate  $\log_3 (27^3)$  using the power rule.

a. 12

b. 9 ✓

c. 11

d. 6

$$\log_3 (27^3) = 3\log_3 27 = 3(3) = 9.$$

**112** Evaluate  $\log_3 (27^2)$  using the power rule.

- a. 5
- b. 4
- c. 8
- d. 6 ✓

$$\log_3 (27^2) = 2\log_3 27 = 2(3) = 6.$$

**113** Solve  $\log_5 (x - 2) = 4$ .

- a. 629
- b. 630
- c. 627 ✓
- d. 625

$$x - 2 = 5^4 = 625 \Rightarrow x = 627.$$

**114** Solve  $\log_4 (x - 1) = 4$ .

- a. 257 ✓
- b. 255
- c. 256
- d. 259

$$x - 1 = 4^4 = 256 \Rightarrow x = 257.$$

**115** Solve  $\log_5 (x) = 3$ .

- a. 123
- b. 128
- c. 125 ✓
- d. 127

$$x = 5^3 = 125.$$

116 Solve  $\log_2(x - 1) = 3$ .

a. 9 ✓

b. 8

c. 7

d. 11

$$x - 1 = 2^3 = 8 \Rightarrow x = 9.$$

117 Solve  $\log_4(x - 2) = 2$ .

a. 21

b. 18 ✓

c. 16

d. 20

$$x - 2 = 4^2 = 16 \Rightarrow x = 18.$$

---

### Multiple choice — hard

118 Solve  $3^x = 80$  for  $x$ , rounded to three decimal places.

a. 4.090

b. 3.890

c. 3.990 ✓

d. 1.903

$$x = \frac{\log 80}{\log 3} \approx \frac{1.903}{0.477} \approx 3.990.$$

**119** Solve  $2^x = 20$  for  $x$ , rounded to three decimal places.

- a. 4.222
- b. 4.422
- c. 4.322 ✓
- d. 1.301

$$x = \frac{\log 20}{\log 2} \approx \frac{1.301}{0.301} \approx 4.322.$$

**120** Solve  $5^x = 50$  for  $x$ , rounded to three decimal places.

- a. 2.531
- b. 2.331
- c. 2.431 ✓
- d. 1.699

$$x = \frac{\log 50}{\log 5} \approx \frac{1.699}{0.699} \approx 2.431.$$

**121** Solve  $3^x = 50$  for  $x$ , rounded to three decimal places.

- a. 3.662
- b. 1.699
- c. 3.562 ✓
- d. 3.462

$$x = \frac{\log 50}{\log 3} \approx \frac{1.699}{0.477} \approx 3.562.$$

**122** Solve  $2^x = 70$  for  $x$ , rounded to three decimal places.

- a. 1.845
- b. 6.130 ✓
- c. 6.030
- d. 6.230

$$x = \frac{\log 70}{\log 2} \approx \frac{1.845}{0.301} \approx 6.130.$$

**123** Solve  $\ln(x) = 3$  for  $x$ , in terms of  $e$ .

a.  $e^3$  ✓

b.  $3e$

c.  $e^4$

d.  $e^2$

$x = e^3$ .

**124** Evaluate  $\ln(e^6)$ .

a. 5

b. 7

c. 6 ✓

d.  $e^6$

$\ln(e^6) = 6$ .

**125** Evaluate  $\ln(e^4)$ .

a. 3

b. 5

c. 4 ✓

d.  $e^4$

$\ln(e^4) = 4$ .

**126** Evaluate  $\ln(e^3)$ .

a. 2

b. 3 ✓

c. 4

d.  $e^3$

$$\ln(e^3) = 3.$$

**127** Evaluate  $\ln(e^2)$ .

a. 1

b.  $e^2$

c. 2 ✓

d. 3

$$\ln(e^2) = 2.$$

**128** A quantity is modeled by  $N(t) = 500e^{-0.1t}$ . Find  $N(5)$ , rounded to one decimal place.

a. 273.0

b. 500

c. 303.3 ✓

d. 333.6

$$N(5) = 500e^{-0.5} = 500(0.60653) \approx 303.3.$$

**129** A quantity is modeled by  $N(t) = 200e^{0.1t}$ . Find  $N(5)$ , rounded to one decimal place.

a. 296.7

b. 200

c. 362.7

d. 329.7 ✓

$$N(5) = 200e^{0.5} = 200(1.64872) \approx 329.7.$$



**130** A quantity is modeled by  $N(t) = 5000e^{0.1t}$ . Find  $N(5)$ , rounded to the nearest whole number.

a. 8244 ✓

b. 5000

c. 9068.4

d. 7419.6

$$N(5) = 5000e^{0.5} = 5000(1.64872) \approx 8244.$$

**131** A quantity is modeled by  $N(t) = 1000e^{-0.1t}$ . Find  $N(5)$ , rounded to one decimal place.

a. 545.9

b. 606.5 ✓

c. 667.1

d. 1000

$$N(5) = 1000e^{-0.5} = 1000(0.60653) \approx 606.5.$$

**132** A quantity is modeled by  $N(t) = 500e^{0.1t}$ . Find  $N(5)$ , rounded to one decimal place.

a. 824.4 ✓

b. 742.0

c. 500

d. 906.8

$$N(5) = 500e^{0.5} = 500(1.64872) \approx 824.4.$$